

Implementation notes: training-free noise-free inpainting of sPHENIX calorimeter images with a DDPM prior

calo-ddpm inpainting study

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1 Setting and notation

Data and model space. Calorimeter events are 24×64 tower- E_T images in the (η, ϕ) plane, in GeV. All sampling operates in the normalized log space $x = \ln(\max(E_T, E_{\min}))$, $E_{\min} = 10^{-3}$ GeV; outputs are mapped back by $E_T = e^x$. The physical log-space range used by all clamps below is $[x_{\min}, x_{\max}] = [-7, 4]$ (≈ 0.9 MeV to 55 GeV).

Diffusion prior. One pre-trained DDPM [1] per centrality class [2], used frozen. Variance-preserving forward process with $T = 8000$ and linear β_t from b/T to $1000 b/T$, $b = 0.02$ (0–10%) or 0.10 (40–50%); the schedule is reconstructed exactly from each checkpoint’s training configuration. With $\bar{\alpha}_t = \prod_{i \leq t} (1 - \beta_i)$ we abbreviate

$$\bar{s}_t = \sqrt{\bar{\alpha}_t}, \quad \bar{v}_t = 1 - \bar{\alpha}_t, \quad (1)$$

stored in arrays of length $S+1$ with an identity transition padded at index 0 ($\bar{s}_0 = 1$, $\bar{v}_0 = 0$). Sampling runs on a subsampled grid $0 = \tau_0 < \tau_1 = 1 < \dots < \tau_S = T$ ($\tau = \text{round}(\text{linspace}(1, T, S))$, $S = 1000$ for the study); the network is always conditioned on the *original* indices τ_k . Below, one reverse step goes from level $s \equiv \tau_k$ to $s-1 \equiv \tau_{k-1}$, with all quantities indexed by the subsampled position.

Denoiser and transport. The ϵ -network gives the Tweedie estimate and the (jump-aware) ancestral step used by all algorithms:

$$\hat{x}_0 = \frac{x_s - \sqrt{\bar{v}_s} \epsilon_\theta(x_s, \tau_k)}{\bar{s}_s}, \quad (2)$$

$$\beta_{(k)} = 1 - \frac{\bar{s}_s^2}{\bar{s}_{s-1}^2}, \quad \text{ddpm_step}(x_s, \hat{x}_0) = \frac{\bar{s}_{s-1} \beta_{(k)}}{\bar{v}_s} \hat{x}_0 + \frac{(\bar{s}_s / \bar{s}_{s-1}) \bar{v}_{s-1}}{\bar{v}_s} x_s + \sqrt{\frac{\beta_{(k)} \bar{v}_{s-1}}{\bar{v}_s}} \xi. \quad (3)$$

At $s = 1$ the padding gives exact collapse, $\text{ddpm_step} = \hat{x}_0$ with zero variance. The chain is initialized at the prior marginal $x_S \sim \mathcal{N}(0, \bar{v}_S \mathbf{I})$. The forward (re-noising) kernels used below are $q_{s-1}(y) = \bar{s}_{s-1} y + \sqrt{\bar{v}_{s-1}} \xi$ (known-content branch) and the one-step jump $x_s = (\bar{s}_s / \bar{s}_{s-1}) x_{s-1} + \sqrt{\beta_{(k)}} \xi$ (RePaint time travel).

Problem. Noise-free masking: binary m ($1 = \text{known/live}$, $0 = \text{dead}$), measurement $y = m \odot x_0^{\text{true}}$, $\sigma_y = 0$; target $p(x_0 \mid m \odot x_0 = y)$. $A = \text{diag}(m)$ is its own pseudoinverse on its range, so no SVD machinery appears anywhere. Posterior sampling batches J samples per truth image ($J = 50$), with K images processed jointly (GPU batch $K \cdot J$); results are stored for the dead region only, in GeV.

2 Projection-type algorithms (calo_inpaint/inpainting/)

2.1 RePaint

Per level s , repeated U times ($U = 10$; a single pass at $s = 1$, where the extra passes of the published Algorithm 1 are deterministic recomputations):

$$x^{\text{known}} = q_{s-1}(y), \quad x^{\text{dead}} = \text{ddpm_step}(x_s, \hat{x}_0), \quad x_{s-1} = m x^{\text{known}} + (1-m) x^{\text{dead}}, \quad (4)$$

with the one-step forward jump between repetitions. The known branch is noised to level $s-1$ (*not* s) so both branches coincide in time after the step — the off-by-one variant biases the dead-region boundary. Output known region equals y exactly ($\bar{s}_0 = 1$, $\bar{v}_0 = 0$).

2.2 DDNM

Range/null-space rectification of the denoised estimate [3], exact for noiseless masking: $\hat{x}_0 \leftarrow m y + (1-m) \hat{x}_0$, followed by Eq. (3). No noisy-case (Σ -scaled) corrections are required. Output equals the projected $\hat{x}_0$ at the last step; known region exact.

2.3 DDRM

The spectral variational posterior of Ref. [4] specializes (pixel basis, $\sigma_y = 0$) to

$$\text{dead: } x_{s-1} = \bar{s}_{s-1} \hat{x}_0 + \sqrt{(1-\eta^2) \bar{v}_{s-1}} \hat{\epsilon} + \eta \sqrt{\bar{v}_{s-1}} \xi, \quad (5)$$

$$\text{known: } x_{s-1} = \bar{s}_{s-1} [(1-\eta_b) \hat{x}_0 + \eta_b y] + \sqrt{\bar{v}_{s-1}} \xi', \quad (6)$$

with $\eta = 0.85$, $\eta_b = 1$ (published defaults) and matching initialization (known pixels start at $q_S(y)$). The known region is additionally projected to y at the final step (a no-op at $\eta_b = 1$). Two properties established for this noise-free limit: (i) the dead-pixel update is the only DDIM-type transport among the implemented algorithms, and its η never passes through the ancestral point ($\eta \rightarrow 1$ degenerates to re-noising $\hat{x}_0$); (ii) with an *exact* denoiser it under-disperses the posterior (std ratio ≈ 0.86 at $\eta = 0.85$; $\rightarrow 1$ as $\eta \rightarrow 0$, ≈ 0.70 at $\eta = 1$; independent of S) — intrinsic to the variational construction, not an implementation artifact.

3 Gradient-guided algorithms (calo_inpaint/guided/)

The gradient-based algorithms in production are adaptations of an implementation developed and validated on CelebA image inpainting. Both follow one template per reverse step; they differ only in the loss argument and the guidance weight. **Bookkeeping note:** these are *not* the published MCG [5] / IIGDM [6] algorithms: both carry a RePaint-style per-step known-region paste (absent from published IIGDM), a clamped transport estimate, and a scalar annealed guidance weight in place of the exact guidance coefficients. They are accurately described as “projection + annealed consistency-gradient guidance” and should be cited as MCG-inspired / IIGDM-inspired variants.

3.1 Common template

1. $\hat{x}_0$ from Eq. (2); clamped copy $\hat{x}_0^c = \text{clip}(\hat{x}_0, x_{\min}, x_{\max})$ used for transport.

2. Consistency loss on the *known* region, per sample (mean over pixel dimensions),

$$\mathcal{L} = \frac{1}{D} \|m \odot (\chi - y)\|_2^2, \quad g = \nabla_{x_s} \mathcal{L} \quad (\text{through the frozen network}), \quad (7)$$

where χ is algorithm-specific (below) and D the pixel count. The squared loss has a smooth gradient everywhere — in particular no 0/0 at zero residual, a failure mode of $\|\cdot\|_2$ -based losses whose autograd gradient is NaN at exactly zero and at overflow.

3. Per-sample gradient clipping, $g \leftarrow g \cdot \min(1, 1/\|g\|_2)$ (per sample, not per batch: a global norm would shrink the guidance with the study batch $K \cdot J$; per-sample preserves the single-image behavior of the original code at any batch size).
4. Guided transport on the dead region and hard paste of the known region at level $s-1$:

$$x^{\text{dead}} = \text{ddpm_step}(x_s, \hat{x}_0^c) - \lambda_s g, \quad x_{s-1} = m q_{s-1}(y) + (1-m) x^{\text{dead}}, \quad (8)$$

with $q_0(y) = y$; a final clean paste follows the loop. The annealed weight λ_s is strong at high noise and vanishes as the chain sharpens.

All gradient evaluations run in fp32 (no autocast), and every step carries finiteness tripwires: a diverging chain raises with the step index rather than writing NaNs to disk.

3.2 MCG-inspired variant (mcg2)

Loss argument $\chi = \hat{x}_0^c$ (clamped) and

$$\lambda_s = g_s (1 - \bar{\alpha}_s), \quad g_s = 0.5. \quad (9)$$

3.3 IIGDM-inspired variant (pigdm2)

Loss argument $\chi = \hat{x}_0$ (unclamped) and a clamped SNR-like pseudoinverse weight,

$$r_s = \min\left(\frac{1 - \bar{\alpha}_s}{\bar{\alpha}_s + 10^{-8}}, r_{\max}\right), \quad \lambda_s = g_s (1 - \bar{\alpha}_s) r_s, \quad g_s = 0.5, \quad r_{\max} = 10. \quad (10)$$

The cap $r_{\max}$ bounds the weight where $\bar{\alpha}_s \rightarrow 0$ (early chain), where the uncapped ratio reaches $\mathcal{O}(10^4)$ on this schedule.

3.4 Stability rationale (lessons from the discarded ports)

Faithful ports of published MCG/IIGDM failed on the trained calorimeter model, and the mechanisms are worth recording:

1. *Amplification.* Eq. (2) multiplies network error by $1/\bar{s}_s$, up to ~ 150 on this schedule; the guidance Jacobian $\partial \hat{x}_0 / \partial x_s$ carries a second such factor. Boundedness of exact-IIGDM-style guidance relies on the near-cancellation $(\mathbf{I} - \sqrt{\bar{v}_s} \partial \hat{\epsilon} / \partial x) / \bar{s}_s$, which holds for an ideal denoiser but not for the trained network (nor under bf16, whose $\sim 0.4\%$ error breaks it outright). Without a per-step projection, the resulting feedback loop diverges within a few steps — reproduced empirically in single precision.
2. *Norm-loss singularities.* Losses of the form $\|r\|_2$ have autograd gradient $r/\|r\|$, which is NaN both at $r = 0$ (exact-zero residual) and at overflow ($\Sigma r^2 \rightarrow \infty$); the squared, per-sample-mean loss of Eq. (7) has neither.

3. *What actually stabilizes the working variant:* the per-step known-region paste bounds the state trajectory, the clamped transport estimate bounds the mean, the per-sample gradient clip bounds the correction, and the annealed λ_s removes guidance precisely where the chain is most error-amplifying. The cost is deviation from the published algorithms, which is why these arms are labeled “-inspired” throughout.

4 Common infrastructure and validation

Table 1: Implemented algorithms and defaults. “Exact y ”: known region of the output equals the measurement identically. NFEs per posterior sample; bwd = additional backward pass per step.

Algorithm	Module	Hyperparameters	Exact y	NFEs
RePaint	<code>inpainting/repaint</code>	$U=10$	yes	$U(S-1)+1$
DDNM	<code>inpainting/ddnm</code>	—	yes	S
DDRM	<code>inpainting/ddrm</code>	$\eta=0.85, \eta_b=1$	yes	S
MCG-inspired	<code>guided/mcg2</code>	$g_s=0.5$	yes	S (+bwd)
IIGDM-inspired	<code>guided/pigdm2</code>	$g_s=0.5, r_{\max}=10$	yes	S (+bwd)

Reproducibility. Each study run draws from dedicated seeded generators, re-seeded per image chunk (seed·1000003+chunk start), so runs are bitwise reproducible for fixed `--images-per-batch` and resume exactly from the last completed chunk. All run parameters are recorded in a `metadata.json` next to the results.

Validation. Three independent checks back the implementations. (i) *Schedule cross-check:* the reconstructed schedule, subsampling, and posterior coefficients agree with the original training code of Ref. [2] to single precision. (ii) *Analytic posterior test:* for an i.i.d. Gaussian prior the optimal ϵ -predictor is closed-form and the dead-pixel posterior is known exactly; all five production algorithms reproduce it to sampling accuracy, and all return the known region exactly. (iii) *Adversarial boundedness:* with a random-weight (worst-case Jacobian) network on the full $T = 8000$ schedule, the guided algorithms remain finite and range-bounded over the entire chain — the configuration under which exact-IIGDM-style guidance diverges.

References

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